

①

a. let:

P: "You have a password"

L: "You can log in"

$$L \Rightarrow P$$

b. let:

F: "You know a formula"

S: "You can solve the problem"

$$F \Rightarrow S$$

2.

P	Q	R	$P \wedge Q$	$(P \wedge Q) \Rightarrow R$	$Q \Rightarrow R$	$P \Rightarrow (Q \Rightarrow R)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	T	T	T
T	F	F	F	T	T	T
F	T	T	F	T	T	T
F	T	F	F	T	F	T
F	F	T	F	T	T	T
F	F	F	F	T	T	T

$$(P \wedge Q) \Rightarrow R \equiv P \Rightarrow (Q \Rightarrow R)$$

③

Standard rule: $A \Rightarrow B \equiv \neg A \vee B$

So, $(P \wedge Q) \Rightarrow R \equiv \neg(P \wedge Q) \vee R$

$P \Rightarrow (Q \Rightarrow R) \equiv \neg P \vee (Q \Rightarrow R)$

For the right hand side:

$Q \Rightarrow R \equiv \neg Q \vee R$

$P \Rightarrow (Q \Rightarrow R) \equiv \neg P \vee (\neg Q \vee R)$

$\neg P \vee (\neg Q \vee R) \equiv \neg P \vee \neg Q \vee R$

Use De Morgan's law

$\neg(P \wedge Q) \vee R \equiv (\neg P \vee \neg Q) \vee R$

We compare both side:

Left = $(\neg P \vee \neg Q) \vee R$

right = $\neg P \vee \neg Q \vee R$

In conclusion: $(P \wedge Q) \Rightarrow R \equiv P \Rightarrow (Q \Rightarrow R)$

(4) a. Base case ($n=5$)

$$2^5 = 32 \text{ and } 5^2 = 25 \quad 32 > 25$$

Induction hypothesis: Assuming for some $k \geq 5$: $2^k > k^2$

Induction step:

$$\text{For hypothesis: } 2^{k+1} = 2 \times 2^k > 2 \cdot k^2$$

So it's enough to prove $2k^2 > (k+1)^2$ for $k \geq 5$

$$2k^2 - (k+1)^2 = 2k^2 - (k^2 + 2k + 1) = k^2 - 2k - 1$$

$$\text{For } k \geq 5, \quad k^2 - 2k - 1 \geq 25 - 10 - 1 = 14 > 0$$

$$\text{Hence } 2k^2 > (k+1)^2, \text{ so } 2^{k+1} > 2k^2 > (k+1)^2$$

b. Base case $n=1$

$$1^3 = 1 = \left(\frac{1 \times 2}{2}\right)^2$$

Induction hypothesis: Assume some $k \geq 1$: $\sum_{i=1}^k i^3 = \left(\frac{k(k+1)}{2}\right)^2$

Induction steps:

$$\sum_{i=1}^{k+1} i^3 = \left(\sum_{i=1}^k i^3\right) + (k+1)^3 = \left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3$$

$$\left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3 = (k+1)^2 \left(\frac{k^2}{4} + k + 1\right) = (k+1)^2 \left(\frac{k^2 + 4k + 4}{4}\right) =$$

$$(k+1)^2 \left(\frac{(k+2)^2}{4}\right) = \left(\frac{(k+1)(k+2)}{2}\right)^2$$



$$\boxed{\text{if } n = k+1, \text{ then } \left(\frac{n(n+1)}{2}\right)^2}$$

c. Base case: $n=1$

$$8^1 - 1 = 7$$

Induction hypothesis: Assume $8^k - 1$ is divisible by 7 for some $k \geq 1$

Induction steps:

If $n=k+1$,

$$8^{k+1} - 1 = 8 \times 8^k - 1 = (7+1)8^k - 1$$

$$8^{k+1} - 1 = 7 \times 8^k + (8^k - 1)$$

By hypothesis that $(8^k - 1)$ can be divisible by 7, we can also prove that $7 \cdot 8^k$ is divisible, then their sum is divisible.

⑤

a. $p \wedge (Q \vee R) \Leftrightarrow (p \wedge Q) \vee (p \wedge R)$

Left side:

Case 1: p is true

$$\text{LHS: } T \wedge (Q \vee R) = (Q \vee R)$$

$$\text{RHS: } (T \wedge Q) \vee (T \wedge R) = (Q \vee R)$$

$$(Q \vee R) = (Q \vee R)$$

Case 2: $(Q \vee R)$ is true

Q is true

$$(p \wedge Q) = \text{True}$$

$$\text{then } (p \wedge Q) \vee (p \wedge R) = T$$

R is true

$$(p \wedge R) = \text{True}$$

$$\text{then, } (p \wedge Q) \vee (p \wedge R) = T$$

Right side:

Case 1: $(p \wedge Q)$ is true

p true and Q true

then $Q \vee R$ true

$$p \wedge (Q \vee R) \text{ true}$$

Case 2: $(p \wedge R)$ is true

p true and R true

then $Q \vee R$ true

$$p \wedge (Q \vee R) \text{ true}$$

b. $P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$

Left:

Case 1: P is true

$P \vee (Q \wedge R)$ is true

$P \vee Q$ is true

$P \vee R$ is true

Case 2: $Q \wedge R$ is true

$P \vee (Q \wedge R)$ is true

$P \vee Q$ is true

$P \vee R$ is true

Right:

Case 1: P true

then:

$P \vee (Q \wedge R)$ is true

$(P \vee Q) \wedge (P \vee R)$ is true

Case 2: $Q \wedge R$ is true

$P \vee (Q \wedge R)$ is true

$(P \vee Q)$ is true

$(P \vee R)$ is true

$(P \vee Q) \wedge (P \vee R)$ is true